

NAME: Sheza Naveed

ROLL NO.: 22L-7651

SECTION: BS-EE (5A)

ASSIGNMENT: ADC 3

QUESTION 1

$$a) m(t) = \sin(2\pi t)$$

$$= \frac{\sin(2\pi t)}{\pi t} \cdot \pi$$

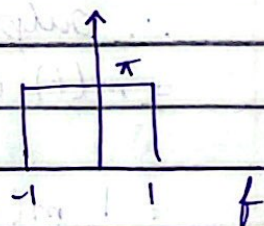
$$s(t) = m(t) \cos(2\pi f_c t)$$
$$= \frac{2 \sin 2\pi t \times \cos(2\pi f_c t)}{2t}$$

$$= \frac{1}{2t} \left[\sin(2\pi(1+f_c)t) + \sin(2\pi(1-f_c)t) \right]$$

$$M(\omega) = \pi * \text{sinc}\left(\frac{\omega}{2\pi}\right)$$

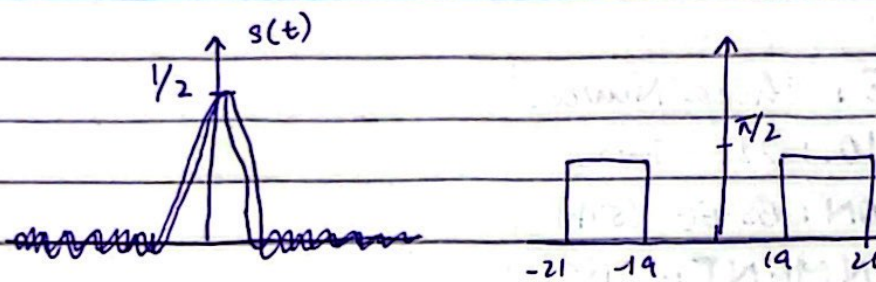
$$M(f) = \pi * \text{sinc}\left(\frac{2\pi f}{2\pi}\right)$$

$$M(f) = \pi * \text{sinc}\left(\frac{f}{2}\right)$$

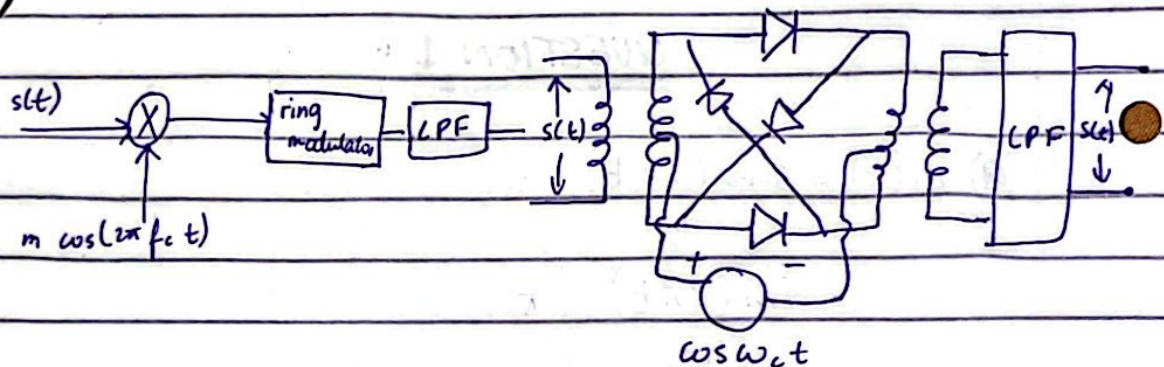


$$s(f) = M(f) * \frac{1}{2} \left[\delta(f-f_c) + \delta(f+f_c) \right]$$

$$= \frac{1}{2} M(f-f_c) + M(f+f_c)$$



b)



Ring Modulator

input signal = $m(t) \cdot \cos 2\pi f_c t$
 - carrier ~~po~~ causes periodic switching on and off of the input signal

$$\therefore \text{output is } m(t) \cos(2\pi f_c t) \cdot x(t)$$

$$= m(t) \cos(\omega_c t) \left[\frac{1}{2} + \frac{2}{\pi} \left(\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \dots \right) \right]$$

$$= \frac{1}{2} m(t) \cos \omega_c t + \frac{2}{\pi} m(t) \cos 2\omega_c t + \dots$$

$$= \frac{2}{\pi} \left[\frac{1 + \cos 2\omega_c t}{2} \right] m(t) + \dots$$

$$= \frac{1}{\pi} m(t) + \underbrace{\frac{m(t) \cos 2\omega_c t}{\pi}}_{\text{eliminated by LPF}} + \dots$$

Use the LPF of gain π

QUESTION 2 :-

Given that

Modulation signal :

$$m(t) = \cos(\omega_m t) + 2 \sin(3\omega_m t)$$

carrier is :

$$c(t) = 3 \cos(\omega_c t) ; \omega_c \gg \omega_m$$

$$\phi_{AM}(t) = (3 + \cos(\omega_m t) + 2 \cos(3\omega_m t)) \cos \omega_c t$$

$$\phi_{AM}(t) = 3 \left(1 + \frac{1}{3} \cos(\omega_m t) \right) + \frac{2}{3} (\cos(3\omega_m t) \cos \omega_c t)$$

$$m_1 = \frac{1}{3} \quad \text{and} \quad m_2 = \frac{2}{3}$$

-total modulation index :

$$m = \sqrt{m_1^2 + m_2^2}$$
$$= \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2}$$

$$m = 0.74$$

therefore, the AM signal ϕ_{AM} satisfy the detection criteria.

Now, we know

$$\phi_{AM}(t) = 3 \left(1 + \frac{1}{3} \cos(\omega_m t) \right) + \frac{2}{3} (\cos(3\omega_m t) \cos \omega_c t)$$

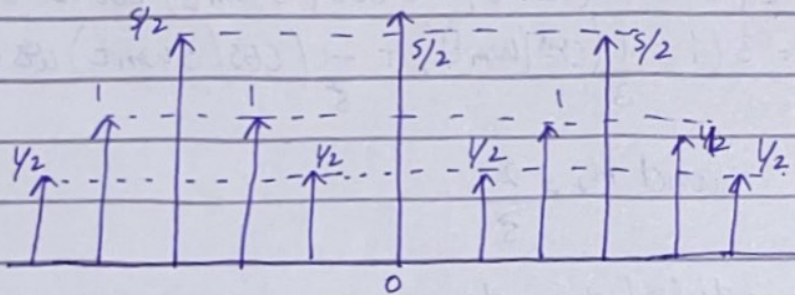
$$= 3 \cos \omega_c t + (\cos \omega_m t \times \cos \omega_c t) + (2 \cos 3\omega_m t) \cos \omega_c t$$

Amplitude of carrier

So,

$$\begin{aligned} \phi_{AM}(\omega) = & \frac{3}{2} (\delta(\omega - \omega_c) + \delta(\omega + \omega_c)) + 1 (\delta(\omega - (\omega_c + \omega_m)) + \delta(\omega + (\omega_c + \omega_m))) \\ & + 1 (\delta(\omega - (\omega_c - \omega_m)) + \delta(\omega + (\omega_c - \omega_m))) + \\ & \frac{1}{2} (\delta(\omega - (\omega_c + 2\omega_m)) + \delta(\omega + (\omega_c + 2\omega_m))) + \frac{1}{2} (\delta(\omega - (\omega_c - 2\omega_m)) + \delta(\omega + (\omega_c - 2\omega_m))) \end{aligned}$$

Spectrum at $\phi_{AM}(\omega)$:



QUESTION 4:

a) We need to analyze the signal at point B, where the baseband signal $m(t)$ is multiplied by $\cos^5(\omega_c t)$. We use binomial expansion for $\cos^5(\omega_c t)$:

$$\cos^5(\omega_c t) = \frac{1}{16} (10 \cos(\omega_c t) + 5 \cos(3\omega_c t) + \cos(5\omega_c t))$$

Now, when $m(t)$ is multiplied by $\cos^5(\omega_c t)$, the signal at point B is:

$$m(t) \cdot \cos^5(\omega_c t) = m(t) \cdot \left[\frac{1}{16} (10 \cos(\omega_c t) + 5 \cos(3\omega_c t) + \cos(5\omega_c t)) \right]$$

this expression represents the modulated signal at point B, where original baseband signal is modulated by a combination of three frequency components: ω_c , $3\omega_c$ and $5\omega_c$.

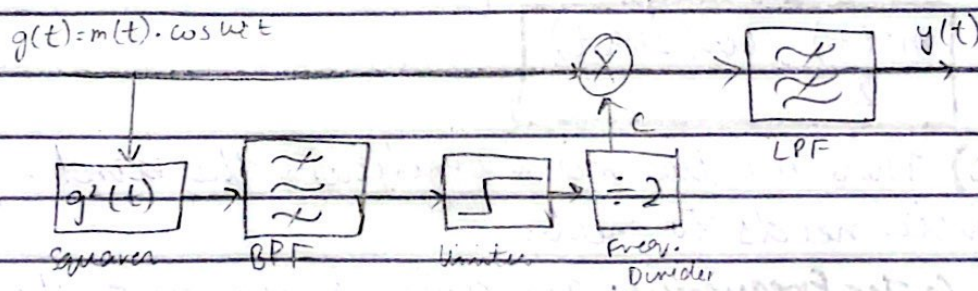
b) At point B, the baseband signal $m(t)$ has been shifted to frequencies around ω_c , $3\omega_c$ and $5\omega_c$, with different amplitudes. The frequency spectra at point B contains components at $\pm \omega_c$, $\pm 3\omega_c$, and $\pm 5\omega_c$, each modulated by a baseband signal spectrum $M(\omega)$.

Thus, after filtering, the signal at point C will be:

$$K m(t) \cos(\omega_c t)$$

This is the desired DSB-SC signal, where K is a constant resulting from the filtering process.

QUESTION 5:-



a) the input signal coming into the square block is $g(t) = m(t) \cdot \cos(\omega_c t)$. Square this signal and find the outcome:

$$g^2(t) = [m(t) \cdot \cos(\omega_c t)]^2$$

$$= m^2(t) \cdot \cos^2(\omega_c t)$$

Now, apply the trigonometric identity

$$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$$

$$g^2(t) = m^2(t) \cdot \frac{1 + \cos(2\omega_c t)}{2}$$

$$g^2(t) = \frac{m^2(t)}{2} \cdot [1 + \cos(2\omega_c t)]$$

b) Point B has a band pass filter, which is tuned for the presence only of the high frequency component, oscillating at $2\omega_c$. It will eliminate the DC component $\frac{m^2(t)}{2}$ while passing only the term that follows:

$$\frac{m^2(t)}{2} \cdot \cos(2\omega_c t)$$

Hence the output at point B remains:

$$\frac{m^2(t)}{2} \cdot \cos(2\omega_c t)$$

c) There are two main conditions the band-pass filter needs to meet:

1. **Center Frequency:** The center frequency of the filter should be adjusted to equal the value of $2\omega_c$ so as not to miss the oscillating component $\cos(2\omega_c t)$.

2. **Bandwidth:** The bandwidth needs to be broad enough to accommodate the modulation in $m(t)$.

to vary the amplitude but at the same time should be narrow enough to reject frequency noise that falls outside of its desired range.

d) the signal is divided in frequency and mixed with original signal to gain oscillation at ω_c .

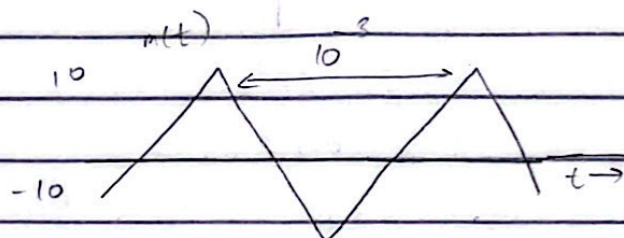
Further, it passes through the low pass filter, due to which high-frequency component is eliminated whereas extracts the original message signal $m(t)$.

Therefore

$$y(t) = m(t)$$

e) The limiter ensures that there is a constant amplitude before the signal gets delivered to the frequency divider. This is necessary because variations in amplitude could distort the signal, therefore, will preserve only the information concerning frequency for proper processing.

QUESTION 6 :-



The AM signal is represented as

a) $S_{AM}(t) = A_c [1 + K_a m(t)] \cos \omega_c t$

where

K_a = amplitude sensitivity

Hence

$m(t)$ is a triangular signal having $|m(t)| = 10$

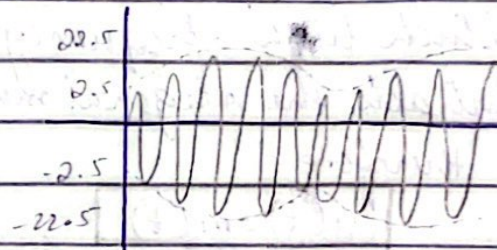
$$|m(t)| = 10 = A_m$$

$$\therefore \mu = \frac{A_m}{A_c} = \text{modulated index}$$

$$\therefore \mu = 0.8$$

$$0.8 = \frac{10}{A_c}$$

$$A_c = 12.5$$

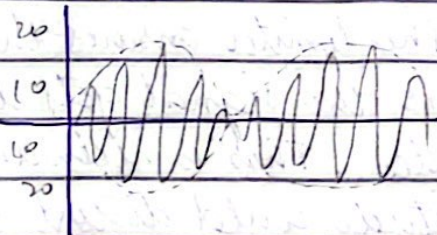


b) $\mu = 1$

$$A_m = 1$$

$$A_c$$

$$A_c = 10$$

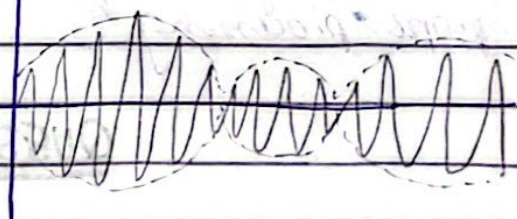


$$\mu = 2$$

$$A_m = 2$$

$$A_c$$

$$A_c = 5$$



a) Given

$$\mu = 0.8$$

$$\therefore \mu = \frac{A_m}{A_c} \Rightarrow 0.8 = \frac{10}{A_c}$$

$$A_c$$

$$A_c$$

$$A_c = 12.5V$$

P_c = carrier ~~power~~ power

$$= \frac{A_c^2}{2}$$

$$= \frac{(12.5)^2}{2}$$

$$\boxed{P_c = 78.125 \text{ W}}$$

b) As we know that $P_t = P_c \left(\frac{1 + \mu^2}{2} \right)$

where

P_t = total AM power and P_c = carrier power

So

$$\frac{P_c \mu^2}{2} = \text{sideband power}$$

Now

$$\text{sideband power} = \frac{78.125 \times (0.8)^2}{2}$$

$$\boxed{P_{SB} = 16.66 \text{ W}}$$

and

$$\begin{aligned} \text{Efficiency } (\eta) &= \frac{P_{SB}}{P_t} \\ &= \frac{P_{SB}}{P_c + P_{SB}} \end{aligned}$$

putting the values

$$\eta = \frac{16.66}{78.125 + 16.66}$$

$$\boxed{\eta = 17.57\%}$$

QUESTION 7:-

a) $b = 8$

$$\begin{aligned}\phi_{AM}(t) &= 2.5 [b + 3 \cos \omega_m t] \cos \omega_c t \\ &= 2.5 [8 + 3 \cos(\omega_m t)] \cdot \cos \omega_c t \\ &= 2.5 \times 8 \left[1 + \frac{3}{8} \cos(\omega_m t)\right] \cdot \cos \omega_c t\end{aligned}$$

$$\text{modulation index} = \frac{3}{8} = 0.375$$

$$\text{power efficiency} = \eta = \frac{\text{sideband power}}{\text{total power}}$$

First we find

$$\text{sideband power} = \frac{\mu^2}{2}$$

$$= \frac{(0.375)^2}{2}$$

$$= 0.070$$

Now,

$$\text{total power} = \frac{1 + \mu^2}{2}$$

$$= \frac{1 + (0.375)^2}{2}$$

$$= 1.070$$

Now putting the values

$$\eta = \frac{0.070}{1.070} \times 100$$

$$\boxed{\eta = 6.569\%}$$

b) $b = 3$

$$\begin{aligned}\phi_{AM}(t) &= 2.5 [b + 3 \cos \omega_m t] \cos \omega_c t \\ &= 2.5 [3 + 3 \cos \omega_m t] \cos \omega_c t \\ &= 2.5 \times 3 [1 + \cos \omega_m t] \cos \omega_c t\end{aligned}$$

modulation index $= 1$

first we find

$$\text{side band power} = \frac{\mu^2}{2}$$

$$= \frac{1}{2} \text{ or } 0.5$$

Now

$$\begin{aligned}\text{total power} &= \frac{1 + \mu^2}{2} \\ &= 1.5\end{aligned}$$

$$\begin{aligned}\text{Efficiency} = \eta &= \frac{P_{SB}}{P_t} \\ &= \frac{0.5}{1.5} \times 100\end{aligned}$$

$$\boxed{\eta = 33.3\%}$$

QUESTION 8:-

Given,

$$m(t) = B \sin(2\pi B t)$$

with

$$B = 1000$$

$$\omega_c = 10000\pi$$

$$a) m(t) = 1000 \operatorname{sinc}(2\pi 1000 t)$$

$$= 1000 \frac{\sin(2000\pi t)}{2000\pi t} ; \operatorname{sinc} \theta = \frac{\sin \theta}{\theta}$$

$$m(t) = \frac{1}{2} \frac{\sin(2000\pi t)}{\pi t}$$

we have

$$\frac{\sin at}{\pi t} \longleftrightarrow \operatorname{rect}\left(\frac{\omega}{2a}\right)$$

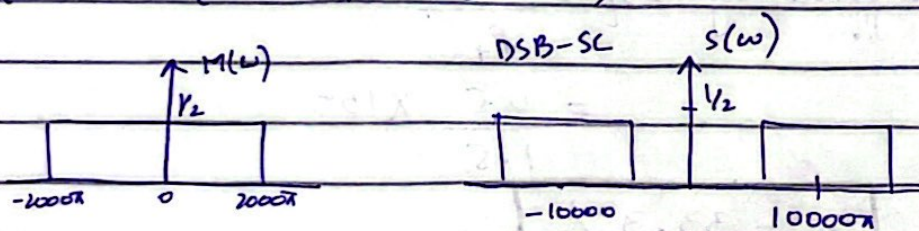
$$M(\omega) = \frac{1}{2} \operatorname{rect}\left(\frac{\omega}{2(2000\pi)}\right)$$

$$s(t) = 2m(t) \cos \omega_c t$$

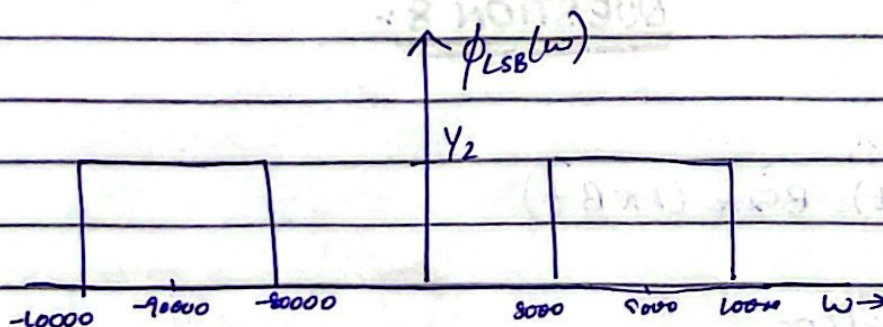
we have

$$x(t) \cos \omega_0 t \longleftrightarrow \frac{1}{2} [x(\omega - \omega_0) + x(\omega + \omega_0)]$$

$$S(\omega) = M(\omega - \omega_c) + M(\omega + \omega_c)$$



b)



$$c) \phi_{LSB}(\omega) = \frac{1}{2} \text{sinc}\left(\frac{\omega - 9000\pi}{2000\pi}\right) + \frac{1}{2} \text{rect}\left(\frac{\omega + 9000\pi}{2000\pi}\right)$$

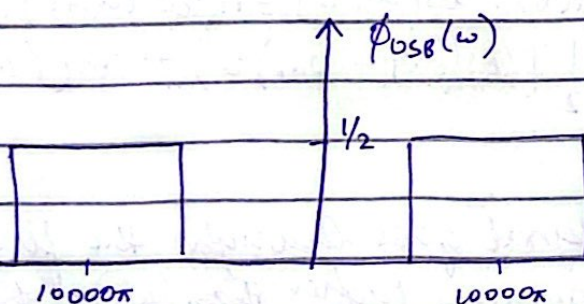
$$\phi_{LSB}(t) = \frac{1}{2} \frac{\sin(1000\pi t)}{\pi t} (2 \cos 9000\pi t)$$

$$\phi_{LSB}(t) = 500 \frac{\sin(1000\pi t)}{1000\pi t} \cdot 2 \cos(9000\pi t)$$

$$\phi_{LSB}(t) = 500 \sin(2\pi 500t) 2 \cos(9000\pi t)$$

$$\boxed{\phi_{LSB}(t) = 1000 \sin(2\pi 500t) \cos(9000\pi t)}$$

d)



$$\phi_{USB}(\omega) = \frac{1}{2} \text{rect}\left(\frac{\omega - 11000\pi}{2000\pi}\right) + \frac{1}{2} \text{sinc}\left(\frac{\omega + 11000\pi}{2000\pi}\right)$$

$$\phi_{USB}(t) = \frac{1}{2} \frac{\sin(1000\pi t)}{\pi t} 2 \cos(11000\pi t)$$

$$= 500 \frac{\sin(1000\pi t)}{\pi t} 2 \cos(11000\pi t)$$

$$\boxed{\phi_{USB}(t) = 1000 \text{sinc}(1000\pi t) \cos(11000\pi t)}$$

QUESTION 9 :-

At Point A :-

At point A, the signal $m(t)$ is modulated with

first oscillator at a frequency of f_1 .
Assuming tone modulation, the modulating signal can be written as

$$m(t) = A \cos(\omega_m t)$$

this has a two sided spectrum centered about $\pm \omega_m$.

The signal is then multiplied by a carrier signal from the first oscillator -

$$\text{Signal at A} = m(t) \cdot \cos(\omega_1 t) = A \cos(\omega_m t) \cos(\omega_1 t)$$

$$\text{signal at A} = \frac{1}{2} [\cos((\omega_1 + \omega_m)t) + \cos((\omega_1 - \omega_m)t)]$$

At Point B:-

When the signal goes through the first BPF which passes only frequencies higher than (ω_1) , the LSB at $(\omega_1 - \omega_m)$ is eliminated.

So, the signal at point B is

$$\text{signal at B} = (\cos(\omega_1 + \omega_m)t)$$

At Point C:-

At point C, the signal is added with the second oscillator at frequency (f_2) . The signal at point B is multiplied by $(\cos(\omega_2 t))$, so the signal at point C becomes

$$[\cos((\omega_1 + \omega_m)t) \cos(\omega_2 t)]$$

Using trigonometric identity,

$$[\cos((\omega_1 + \omega_m)t) \cos(\omega_2 t) = \frac{1}{2} [\cos((\omega_1 + \omega_m + \omega_2)t) + \cos((\omega_1 + \omega_m - \omega_2)t)]]$$

At Point D :

The second BPF filters the output of the mixture and allows only the higher frequency component which represent the upper side band -

So,

$$[\cos(\omega_1 t + \omega_{mt} + \omega_2 t)]$$

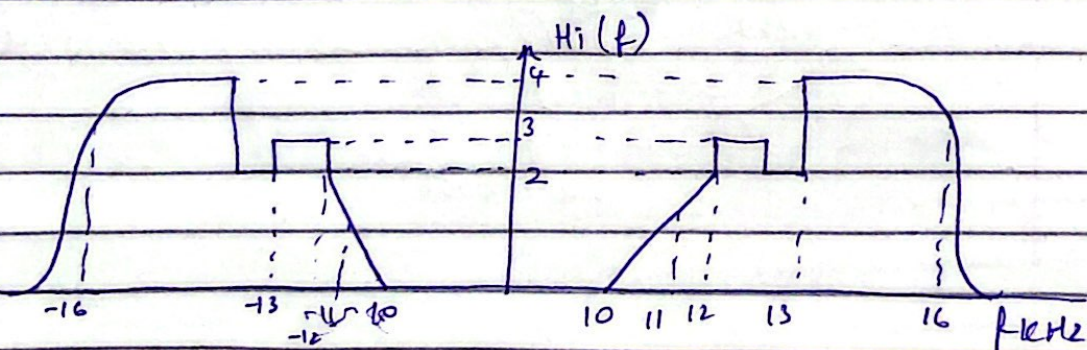
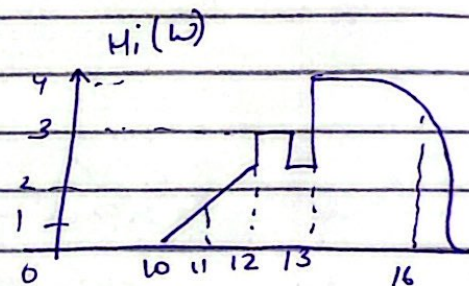
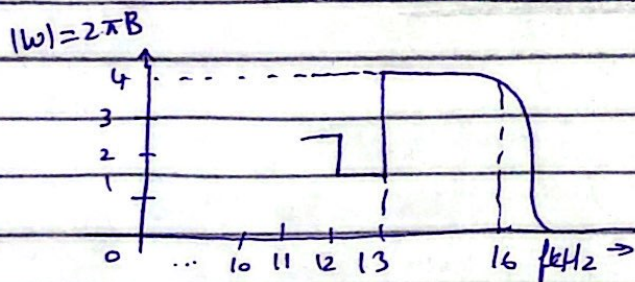
This is the end of the last USB signal which has the carrier frequency ($f_c = f_1 + f_2$)

QUESTION 10 :

Given $f_c = 11 \text{ kHz}$
 $B = 5 \text{ kHz}$

we have

$$H_o(\omega) = \frac{1}{H_i(\omega + \omega_c) + H_i(\omega - \omega_c)}$$



$H_i(f + f_c)$ is $(-f_c)$ centered version of $H_i(f)$

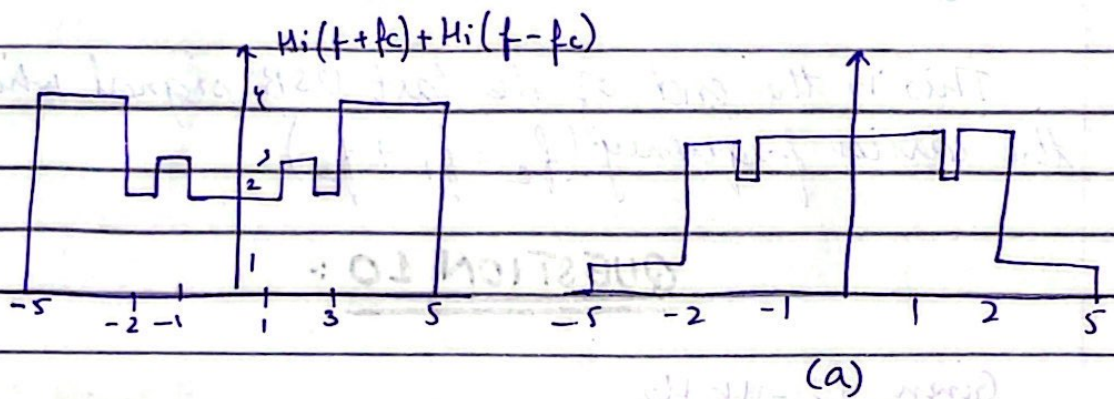
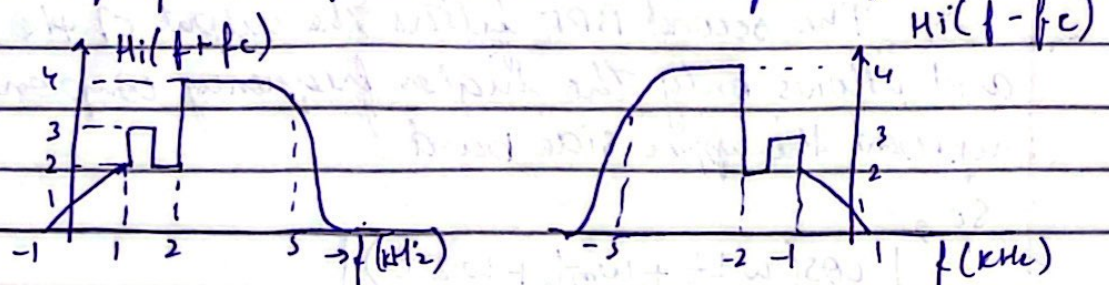
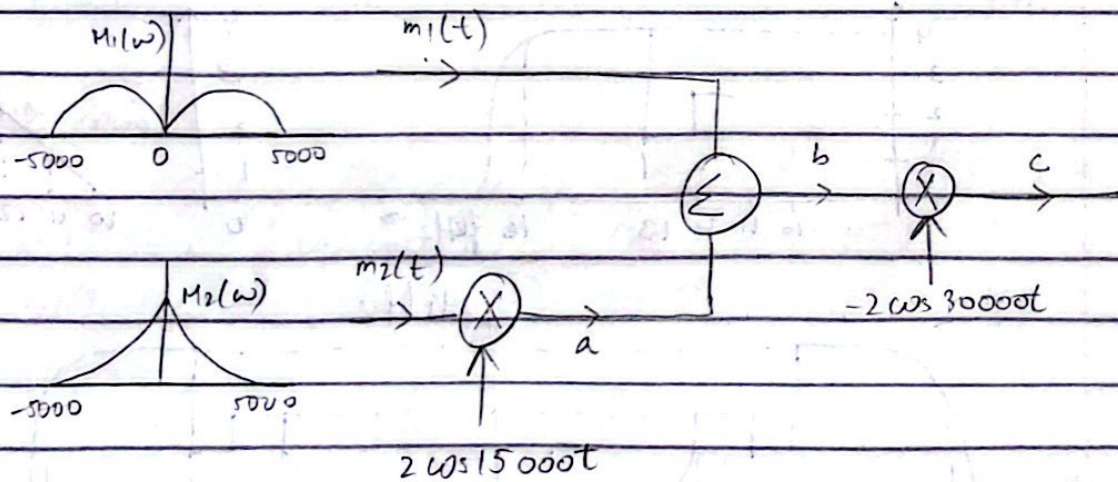


Figure (a) is $H_o(f) = \frac{1}{H_i(f + f_c) + H_i(f - f_c)} : |f| \leq B$

QUESTION 3:-



a) Point a:

The signal ($m_2(t)$) is modulated with the carrier whose frequency (15000 rad/s) at point a. Since ($m_2(t)$) is band limited to ($\pm 5000 \text{ rad/s}$), its frequency spectrum will be shifted and will appear between (10000 rad/s) and (20000 rad/s).

Point b:

At point b, we sum the modulated ($m_2(t)$) and the signal ($m_1(t)$). The frequency spectrum will have two components: That is, the spectrum of ($m_1(t)$) will be band limited from (-5000 rad/s) $\rightarrow$ (5000 rad/s).

$m_2(t)$ signal occupies the frequency range (10000 rad/s) to (20000 rad/s) and is modulated.

Point c:

The combined signal is modulated at point c with a carrier of (30000 rad/s). The spectrum will be shifted as follows.

$m_1(t)$ is shifted from (25000 rad/s) $\rightarrow$ (35000 rad/s).

Instead (40000 rad/s) $\rightarrow$ (60000 rad/s) will be shifted to the modulated ($m_2(t)$) signal.

b) At point c, the total bandwidth required is the difference between the highest and lowest frequency components of the modulated signal.

$m_1(t)$ is (25000 rad/s) to (35000 rad/s) (10000 rad/s) to (40000 rad/s) (10000 rad/s) occupy $m_2(t)$.

thus, total bandwidth is

$$\text{bandwidth} = 25000 \text{ rad/s} - 50000 \text{ rad/s}$$

$$B = 25000 \text{ rad/s}$$

c) To recover the original signal $m_1(t)$ and $m_2(t)$ at point c, we use the following steps:

1. For $m_1(t)$:

A band pass filter is implemented to isolate the $m_1(t)$ signal and pass frequencies between (25000 rad/s) and (35000 rad/s). This signal is demodulated by mixing with a (30000 rad/s) carrier to bring it back to baseband.

2. For $m_2(t)$:

Isolate the signal ($m_2(t)$) with a BPF that passes frequencies (40000 rad/s) to (50000 rad/s). Then demodulate it again with a carrier of (30000 rad/s) and then demodulate again with (15000 rad/s) to get it back to baseband.